CS661: Project Proposal Report
OncoLens: A High-Performance Visual Analytics System for Multi-Dimensional Brain Tumor Characterization

GROUP-11

1 Introduction

Brain cancer encompasses a diverse and highly complex group of central nervous system malignancies. Differentiating between distinct clinical pathologies is a fundamental challenge in computational biology and precision oncology. Tumors that look histologically similar under a microscope can behave in vastly different patterns at the molecular level, causing significant variances in therapeutic responses and patient survival rates.

Our project aims to develop **OncoLens**, an interactive web-based visual analytics system designed for multi-dimensional brain tumor characterization and the genomic dysregulation mapping. Using the curated CuMiDa brain cancer gene expression dataset, our system will compress thousands of genetic dimensions down into an intuitive visual application. The primary objective of the system is to visually map out, analyze, and decode the subtle molecular and structural boundaries that separate healthy tissue from four aggressive forms of the brain cancer: *Ependymoma, Glioblastoma, Medulloblastoma, and Pilocytic Astrocytoma.*

2 Data Source

The primary data source for our project is the Brain Cancer Gene Expression Dataset from the CuMiDa (GSE50161) repository available on Kaggle. The dataset contains high-quality transcriptomic profiles spanning the following structural dimensions:

- **Malignancy Classifications:** 130 distinct patient tissue profiles categorized into five clinical classes : *Ependymoma*, *Glioblastoma*, *Medulloblastoma*, *Pilocytic Astrocytoma*, and *Normal Brain Tissue.*
- **Expression Values:** 54,676 unique genetic microarray probe profiles (columns) recording continuous numerical signal intensities tracking cellular activity shifts.

3 Specific Tasks

To handle the massive dimensionality of this dataset without causing browser latency, the application backend will execute several data processing and mathematical derivation pipelines upon initialization:

- **Variance Threshold Filtering:** Compute the statistical variance of all 54,676 features across all samples using `pandas`. The system will drop flatlining or uninformative features and isolate the top 1,000 highly-varying oncogenes to preserve runtime performance.

- **Genomic Coordinate Integration:** Cross-reference the remaining microarray probe identifiers with an external public genomic assembly reference table to derive three new spatial attributes: *Chromosome Number*, *Base-Pair Start Position*, and precise *Cytoband Location* (e.g., *19q13.3*).
- **Data Transformation & Matrix Calculations:** Derive a continuous patient-specific metric called the **Dysregulation Index (DI)** using standard scaling to evaluate how heavily a tumor's molecular footprint deviates from a healthy cell baseline:

$$DI_p = \sum_{g \in \text{Top Genes}} \left| \frac{x_{pg} - \mu_{\text{Normal},g}}{\sigma_{\text{Normal},g}} \right| \tag{1}$$

  where x_{pg} represents the expression level of gene g for patient p, while $\mu_{\text{Normal},g}$ and $\sigma_{\text{Normal},g}$ represent the historical mean and standard deviation of that gene across healthy baseline controls.
- **User Interaction Optimization:** Develop intuitive visualization tools that allow users to highlight subsets of data, explore gene-specific expression profiles, and compare patterns across different brain cancer subtypes.

4 Visualization Tasks

We will implement five specific, highly interactive visual analytics tasks to uncover hidden biological patterns and present a data-driven narrative:

4.1 Chromosomal Hotspot & Cytoband Mapping

Understanding where genetic modifications physically cluster within human DNA tracks is essential for identifying large-scale structural variations. This visualization interface will allow users to perform the following core analytical functions:

- **Macro-Level Assembly Exploration:** View a global, synchronized physical assembly map representing human chromosomes positioned side-by-side.
- **Locus Deviation Profiling:** Map individual genetic loci coordinates (base-pair start positions along the genomic sequence) against their derived, continuous *Dysregulation Index* metrics.

Suggested Visualization: A high-performance, multi-track linear facet plot built using the `plotly.express.scatter` mapped with the parameter `facet_col='Chromosome'`. The layout engine will embed reactive callback actions to handle point selections; when a user clicks on an active chromosome track, the application dynamically updates the coordinate space with an animated zoom transition, focusing down onto precise sub-chromosomal cytobands to isolate dense malignant hotspots.

4.2 Subtype Volumetric Biomarker Profiling

Evaluating how specific oncogenes vary across different pathologies is a critical step in isolating diagnostic baselines and establishing target signatures. This panel allows researchers to evaluate single-gene behavior through the following capabilities:

- **Dynamic Cohort Querying:** Search and filter through the top variable genes on the fly using an integrated, autocomplete-enabled text query and dropdown interface.

- **Statistical Profiling:** Instantly observe and benchmark mathematical expression distributions, median lines, quartiles, and statistical variance spreads across the different tissue groups.

Suggested Visualization: A split side-by-side Violin-Plot and Box-Plot panel using the `plotly.express.violin` configured with `box=True` and `points="all"`. The graphic environment automatically overrides and redraws its geometry layouts in real time based on user gene selections.

4.3 Multi-Gene Phenotypic "Joystick" Contour Plot

Malignancy is an emergent property driven by a multi-variable genetic ecosystem rather than a single isolated gene. This visualization helps analyze how multi-gene expression thresholds shift or overlap to create distinct diagnostic boundaries. Users can:

- **Axis Parameter Customization:** Assign separate individual genes to the X and Y dimensions independently using dual interactive dropdown selectors.
- **Boundary Overlap Evaluation:** Evaluate topographical and spatial density loops to identify where patient groups from different classifications intersect or segregate.

Suggested Visualization: A 2D Contour Density Landfill chart generated through using the `plotly.express.density_contour`. The backend execution pipeline computes 2D kernel density estimations (KDE) to smoothly draw color-filled probability contours, highlighting the phenotypic thresholds where healthy baseline tissue transitions into malignant clusters.

4.4 Gene Interaction & Co-occurrence Network

This module tracks how functional biological pathways reorganize, strengthen, or completely fracture inside deadly tumors compared to healthy baseline controls, mapping true cellular network disruption. It will allow users to:

- **Correlation Thresholding:** Explore high-confidence co-expression links based on Pearson correlation values ($r > 0.7$) calculated dynamically across target genes.
- **Pathology-Specific Subsetting:** Filter, slice, and toggle network node arrangements by individual cancer subtypes to isolate disease-specific network behaviors.

Suggested Visualization: A custom 2D node-link graph visualization powered by a backend `NetworkX` layout generator and rendered via Plotly graphics primitives. Individual nodes map specific genes, while connecting edge thicknesses correspond directly to the calculated strength of the correlation.

4.5 Interactive Virtual Expression Assayer (The Simulator Widget)

To demystify "black-box" clinical predictive models, this specialized widget provides a real-time, exploratory sandbox environment for simulation. Users can directly interface with the data framework to:

- **Expression Footprint Manipulation:** Adjust continuous expression parameters across three core diagnostic marker genes simultaneously using manual slider controls.
- **Real-Time Class Transitioning:** Observe live classification probability adjustments computed dynamically by an underlying distance classifier.

Suggested Visualization: A horizontal multi-bar chart layout mapping cumulative classification probabilities (0% to 100%) across the five tissue pathologies. The backend applies a Softmax normalization across calculated Euclidean distance profiles to the target centroids, ensuring the visual bars smoothly animate left and right as input sliders change.

5 Tech Stack

- **Backend & Analytics:** Python, `Pandas` and `NumPy` for matrix filtering, `SciPy` for KDE calculations, and `scikit-learn` for distance-based probability modeling.
- **Frontend Framework:** Python `Dash` (built on React.js and Flask) for responsive web applications, integrated with **D3.js** to enable dynamic and visually engaging exploration of genomic and clinical data.
- **Visualization Engine:** `Plotly` for high-performance, WebGL-accelerated interactive renderings.

6 Team Members & Responsibilities

Each member will contribute to different aspects of the project to ensure the successful development of **OncoLens**, covering data processing, analytical modeling, visualization design, frontend development, and system integration.

- **Data Processing & Genomic Annotation:** Krishna Khetre (240543), Ayush Dwivedi (240238)
- **Backend Development & Analytical Modeling:** Sameer Baranwal (240921), Ashma Singh (240215)
- **Visualization Development:** Shantanu Chhonkar (240960), Samarth Awasthi (240917)
- **Frontend Development & User Interaction:** Dhruv Paliwal (240358), Adithya Vishnu (240047)

This project aims to build a comprehensive visual analytics platform for multi-dimensional brain tumor characterization by combining large-scale genomic data processing, statistical analysis, interactive visualizations, and user-centric exploratory tools. The responsibilities have been distributed to ensure complete coverage of all major components required for the successful development and deployment of the proposed system.